

# RCBD

- Most widely used experimental design in field research
- A primary distinguishing feature of RCBD from CRD – the presence of **blocks** of equal size
  - Each block contains all treatments
- Purpose of blocking – to reduce experimental error
  - By eliminating a known source of variation among experimental units other than treatment

# RCBD

- Experimental units – grouped into blocks according to known or suspected source of variation
- Known source of variation
  - May exist in the form of a gradient – in one direction
  - Can be isolated by the blocks
- Conditions – source of variation
  - As homogenous as possible within blocks
  - May differ greatly between blocks
- **Each block has all treatments**
  - The source of variation other than treatment can be corrected

# RCBD

- Blocks made
  - Treatments assigned at random within each block
  - Complete randomization within a block
  - Each block with a separate randomization
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- One of the most widely used designs in field
  - Flexible design – any number of treatments and blocks possible

# RCBD

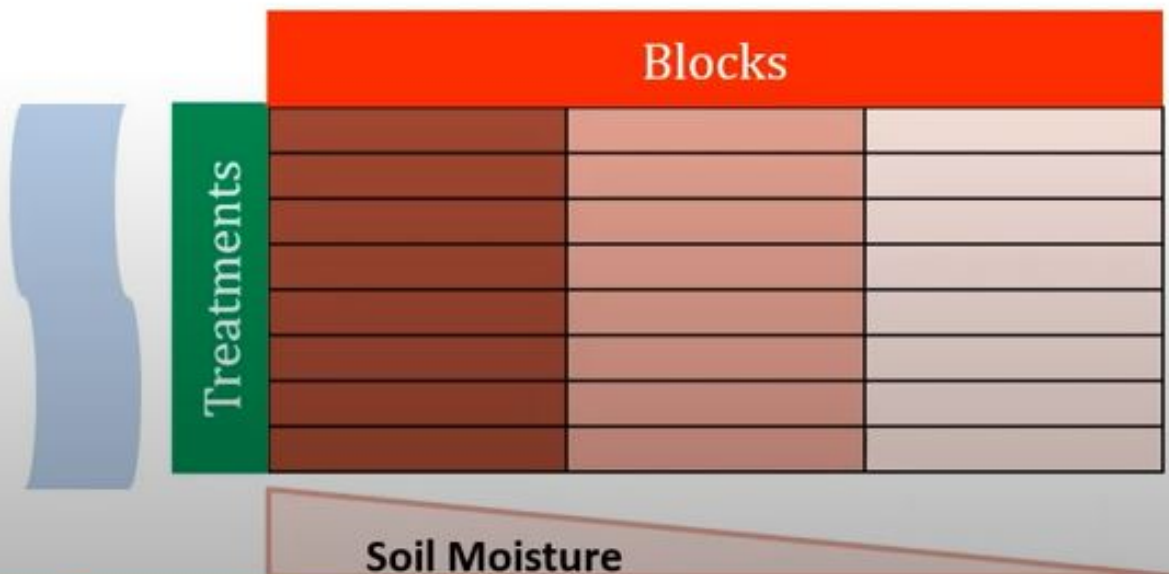
- Long and narrow blocks – designed
- Length of blocks – perpendicular to the direction of gradient
- If there are two sources of variation that occur in a gradient – use Latin square design

# RCBD

- Example
- Effects of fertilizer dose on the fresh weight of plants
- Experimental design – RCBD
- Independent variable – fertilizer doses
- Treatments – different doses of fertilizer (A-H)
- Dependent (response) variable – fresh weight of plants

# RCBD

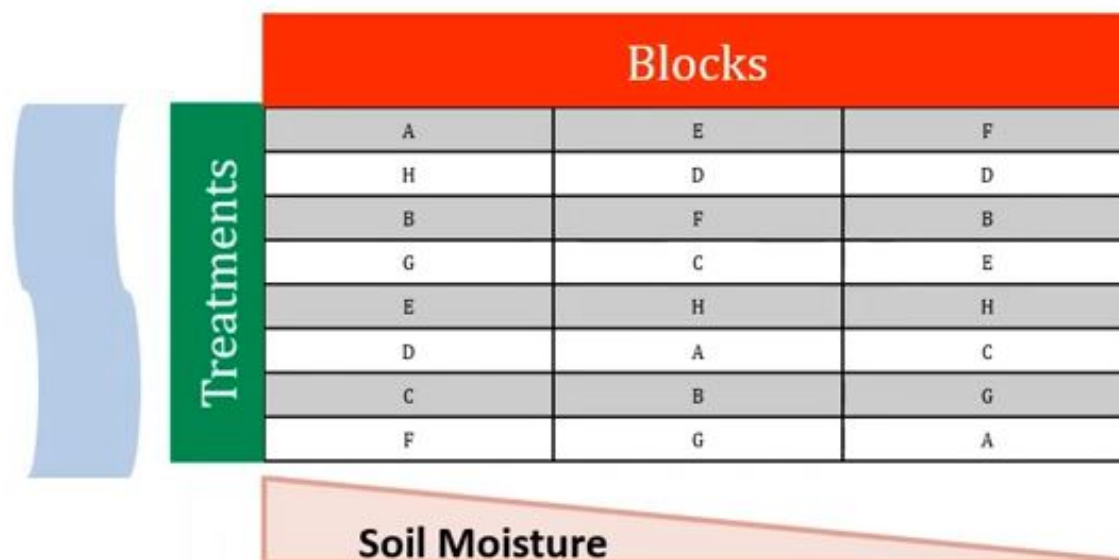
- **Fresh weight of plants** is measured after 6 months of planting to see the **effect of fertilizer** in field trial
- There is **river** on one side – soil near river more moist – decreases as we move away from it



Within each block

- Soil moisture effect – kept constant
- Treatments vary

# RCBD



- **3 blocks** – eliminate the river gradient (a source of variation)
- **8 treatments** (our source of variation of interest)
- Each block has all treatments
- Within each block – complete randomization of treatments

# RCBD

- Can be effectively analyzed through ANOVA
- **ANOVA**
- Three sources of variability in RCBD
  - **Treatment** – variation of interest
  - **Block** – unavoidable variation
  - **Experimental error**
- $\text{Variability}_{\text{Total}} = \text{Variability}_{\text{Treatment}} + \text{Variability}_{\text{Block}} + \text{Variability}_{\text{ExperimentalError}}$
- One more than CRD

# RCBD

- Fresh weight (g) of plants under different doses of fertilizer (treatments A-H)

| Treatments  | Blocks |        |        | Treatment total                        | Treatment mean      |
|-------------|--------|--------|--------|----------------------------------------|---------------------|
|             | I      | II     | III    |                                        |                     |
| A           | 30.85  | 38.01  | 35.10  | 103.96                                 | 34.65               |
| B           | 30.24  | 28.43  | 35.93  | 94.60                                  | 31.53               |
| C           | 30.94  | 31.64  | 34.95  | 97.53                                  | 32.51               |
| D           | 29.89  | 29.12  | 36.75  | 95.76                                  | 31.92               |
| E           | 21.52  | 24.07  | 20.76  | 66.35                                  | 22.12               |
| F           | 25.38  | 32.14  | 32.19  | 89.71                                  | 29.90               |
| G           | 22.89  | 19.66  | 26.92  | 69.47                                  | 23.16               |
| H           | 29.44  | 24.95  | 37.99  | 92.38                                  | 30.79               |
| Block total | 221.15 | 228.02 | 260.59 | Grand total<br>$\sum x_{Tj}$<br>709.76 | Grand mean<br>29.57 |

# RCBD

- $SS_{\text{Total}} = SS_{\text{Treatment}} + SS_{\text{Blocks}} + SS_{\text{ExperimentalError}}$
- For the sake of simplicity
- $SS_{T_o} = SS_t + SS_b + SS_e$
- **Calculate CT**
- $CT = \frac{(\sum x_{T_o})^2}{(n_b \times n_t)} = \frac{(709.76)^2}{3 \times 8}$
- **CT = 20989.97**

# RCBD

- Calculate  $SS_{T_0}$
- $SS_{T_0} = \sum x_{T_0}^2 - CT$
- $SS_{T_0} = [(30.85)^2 + (38.01)^2 + \dots + (37.99)^2] - 20989.97$
- **$SS_{T_0} = 678.42$**
- Calculate  $SS_t$
- $SS_t = \{[(\sum x_{t1})^2/n_{t1}] + [(\sum x_{t2})^2/n_{t2}] + \dots + [(\sum x_{t8})^2/n_{t8}]\} - CT$
- $SS_t = \{[103.96)^2/3] + [(94.60)^2/3] + \dots + [(92.38)^2/3]\} - 20989.97$
- **$SS_t = 426.45$**

# RCBD

- **Calculate  $SS_b$**

- $SS_b = \{[(\sum x_{b1})^2 / n_{b1}] + [(\sum x_{b2})^2 / n_{b2}] + [(\sum x_{b3})^2 / n_{b3}]\} - CT$

- $SS_b = \{[(221.15)^2 / 8] + [(228.02)^2 / 8] + [(260.59)^2 / 8]\} - 20989.97$

- **$SS_b = 110.98$**

- **Calculate  $SS_e$**

- $SS_{T0} = SS_t + SS_b + SS_e$

- $SS_e = SS_{T0} - (SS_b + SS_t)$

- $SS_e = 678.42 - (110.98 + 426.45)$

- **$SS_e = 140.98$**

# RCBD

- Calculate df
- Total df =  $(b \times t) - 1 = (3 \times 8) - 1 = 24 - 1 = 23$
- Treatment df =  $t - 1 = 8 - 1 = 7$
- Block df =  $b - 1 = 3 - 1 = 2$
- Error df =  $(b - 1)(t - 1)$   
 $= (3 - 1)(8 - 1) = 2 \times 7 = 14$

# RCBD

- **Calculate MS**

- $MS = SS/df$

- $MS_t = SS_t / df_t$   
 $= 426.45 / 7 = \mathbf{60.92}$

- $MS_b = SS_b / df_b$   
 $= 110.98 / 2 = \mathbf{55.49}$

- $MS_e = SS_e / df_e$   
 $= 140.98 / 14 = \mathbf{10.07}$

# RCBD

- Calculate F

- $F_t = MS_t / MS_e$   
 $= 60.92 / 10.07 = 6.05$

- $F_b = MS_b / MS_e$   
 $= 55.49 / 10.07 = 5.51$

# RCBD

- **ANOVA table**

| Source of variation | SS     | df | MS    | F    | P      |
|---------------------|--------|----|-------|------|--------|
| Treatment           | 426.45 | 7  | 60.92 | 6.05 | < 0.01 |
| Block               | 110.98 | 2  | 55.49 | 5.51 | < 0.05 |
| Error               | 140.98 | 14 | 10.07 |      |        |
| Total               | 678.42 | 23 |       |      |        |

- Tabled F value for treatment = **2.76** at p 0.05; **4.28** at p 0.01; **7.08** at p 0.001
- Tabled F value for block = **3.74** at p 0.05; **6.52** at p 0.01; **11.78** at p 0.001

# RCBD

- **Result**
- There is highly significant effect of fertilizer on the fresh weight of plants ( $F_{7,14} = 6.04$ ,  $p < 0.01$ ).
- There is significant effect of river on the fresh weight of plants ( $F_{2,14} = 5.51$ ,  $p < 0.05$ ).